

Program : Diploma in Artificial Intelligence And Engineering	
Course Code : 3381	Course Title: Ethics in Artificial Intelligence
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives

- Provide a comprehensive introduction to the ethics in artificial intelligence

Course Prerequisites:

Nil

Course Outcomes :

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the impact of AI	12	Understanding
CO2	Recognise and address ethical issues in AI and understand bias mitigation strategies	11	Understanding
CO3	Demonstrate the ability to address ethical concerns, ensure transparency, and mitigate biases and harms associated with AI systems and datasets.	10	Understanding
CO4	Understand ethical challenges and opportunities in AI	10	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2				2		
CO2	2				2		
CO3	2				2		
CO4	2				2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the impact of AI		
M1.01	Define the morality and ethics in AI	2	Understanding.
M1.02	Outline the impact of AI on society	2	Understanding.
M1.03	Outline the impact of AI on human psychology	2	Understanding.
M1.04	Outline the impact of AI on legal system	2	Understanding.
M1.05	Outline the impact of AI on environment & the planet	2	Understanding
M1.06	Outline the impact of AI on trust	2	Understanding
Contents: Definition of morality and ethics in AI- Impact on society-Impact on human psychology- Impact on the legal system -Impact on the environment and the planet-Impact on trust.			
CO2	Recognise and address ethical issues in AI and understand bias mitigation strategies		
M2.01	Understand International Ethical Initiatives	3	Understanding
M2.02	Identify Ethical Harms and Concerns	3	Understanding
M2.03	Analyze Bias in AI Systems	2	Understanding
M2.04	Summarize Bias Mitigation Techniques	3	Understanding

	Series Test – I	1	
Contents: Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms			
CO3	Demonstrate the ability to address ethical concerns, ensure transparency, and mitigate biases and harms associated with AI systems and datasets.		
M3.01	Apply ethical considerations throughout AI system design.	2	Understanding
M3.02	Understand transparency in autonomous systems.	2	Understanding
M3.03	Understand the role of ontological standards in ethically driven robotics.	2	Understanding
M3.04	Identify challenges to government adoption of AI	2	Understanding
M3.05	Understand the role of datasets and mitigate associated harms.	2	Understanding
Contents: AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design – Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms.			
CO4	Understand ethical challenges and opportunities in AI		
M4.01	Identify ethical challenges and opportunities in AI.	3	Remembering
M4.02	Analyze societal implications of AI applications	3	Understanding
M4.03	Understand decision-making roles across industries in the context of AI.	2	Understanding
M4.04	Understand national and international strategies for governing AI.	2	Understanding
	Series Test – II	1	

Contents:**AI AND ETHICS- CHALLENGES AND OPPORTUNITIES**

Challenges – Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI.

Text / Reference

T/R	Book Title/Author
T1	Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 – March 2020
R1	Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020
R2	Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/106/105/106105214/#
2	https://www.tutorialspoint.com/operating_system/index.htm